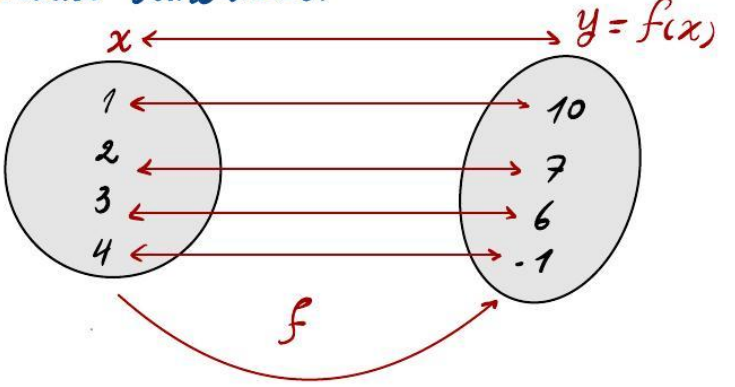
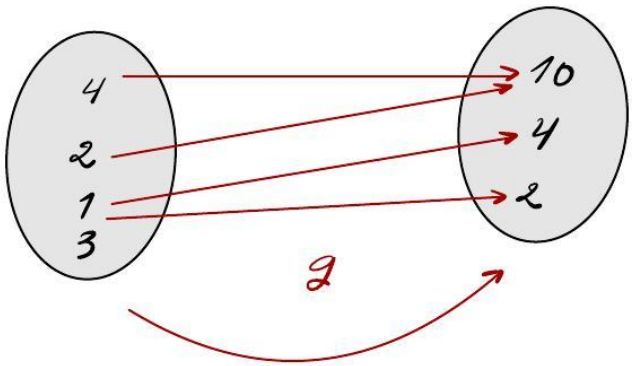
Lecture 1. Jan 20 - 2026.

6.1. Inverse functions and Their Derivatives.

* Inverse functions.





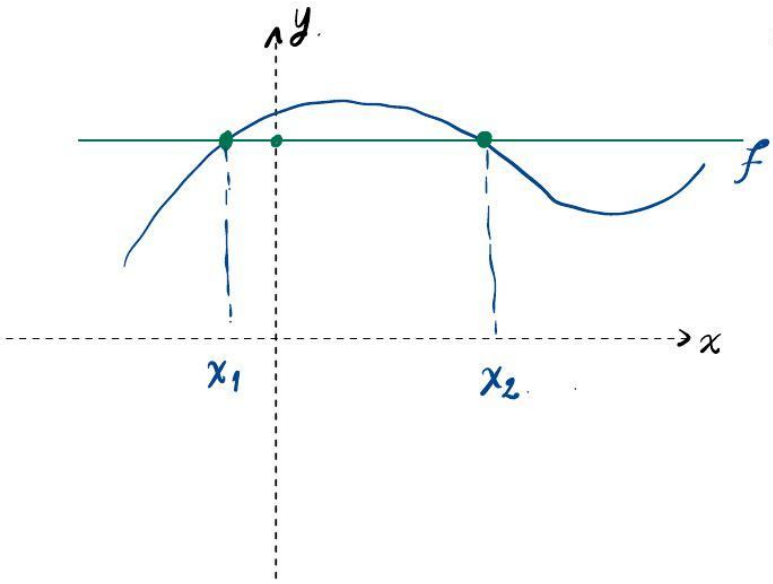
Not all functions possess inverses.

Def A function f is called a **one-to-one** function if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever} \quad x_1 \neq x_2.$$

Horizontal line test

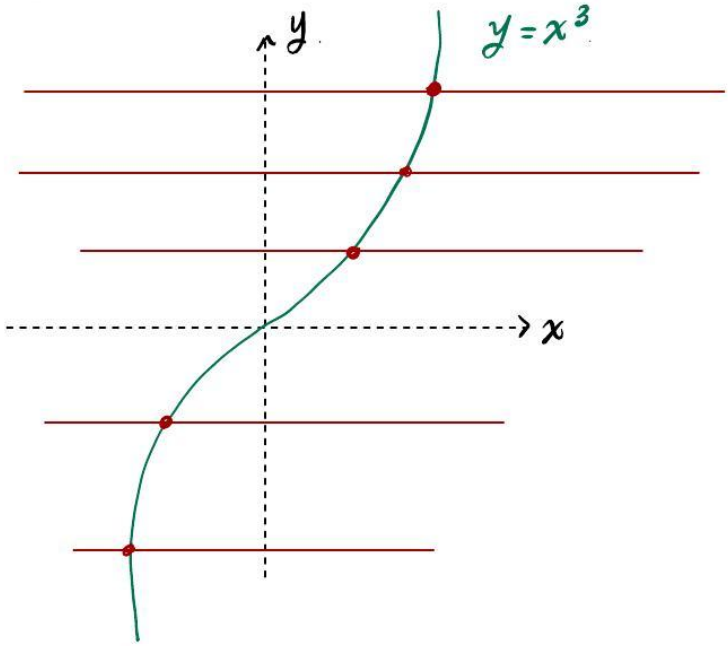
A function is one-to-one if and only if no horizontal line intersects its graph more than once.



Example. is the function $f(x) = x^3$ one-to-one?

Sln. if $x_1 \neq x_2$ then $x_1^3 \neq x_2^3$ (two different numbers can't have the same cube). Therefore, by Definition $f(x) = x^3$ is one-to-one.

Sln. We know.



We see that no horizontal line intersects the graph of $f(x) = x^3$ more than once. Therefore, by Horizontal line test, f is one-to-one.

Example: is the function $g(x) = x^2$ one-to-one?

Sln. This function is not one-to-one b/c.

$x_1 = 2$, $x_2 = -2$ are different but

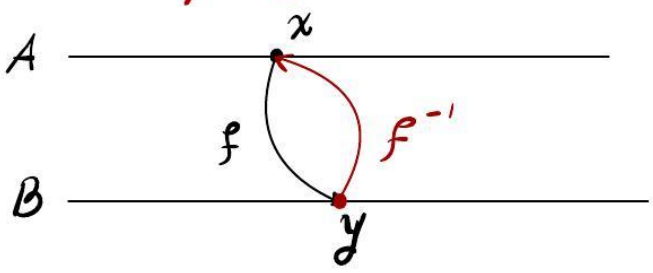
$$f(x_1) = (2)^2 = 4 \quad \text{and} \quad f(x_2) = (-2)^2 = 4$$

i.e, $f(x_1) = f(x_2)$

Def Let f be a one-to-one function with domain A and range B. then its inverse function f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \quad \text{iff} \quad f(x) = y.$$

Rmk. This definition says that if f maps x into y, then f^{-1} maps y back into x.



$$\text{Dom } f^{-1} = \text{Range } f$$

$$\text{Range } f^{-1} = \text{Dom } f$$

for example, the inverse function of $f(x)=x^3$ is $f^{-1}(x)=x^{1/3}$ because if $y=x^3$ then

$$f^{-1}(y)=f^{-1}(x^3)=(x^3)^{1/3}=x$$

⚠ Do not mistake the -1 in f^{-1} for an exponent. Thus,

$f^{-1}(x)$ does not mean $\dfrac{1}{f(x)}$.

Example If f is one-to-one function and $f(1)=5$, $f(3)=7$, $f(8)=-10$ find $f^{-1}(7)$, $f^{-1}(5)$, $f^{-1}(-10)$.

Sln.

$f^{-1}(7)=3$ b/c $f(3)=7$

$f^{-1}(5)=1$ b/c $f(1)=5$

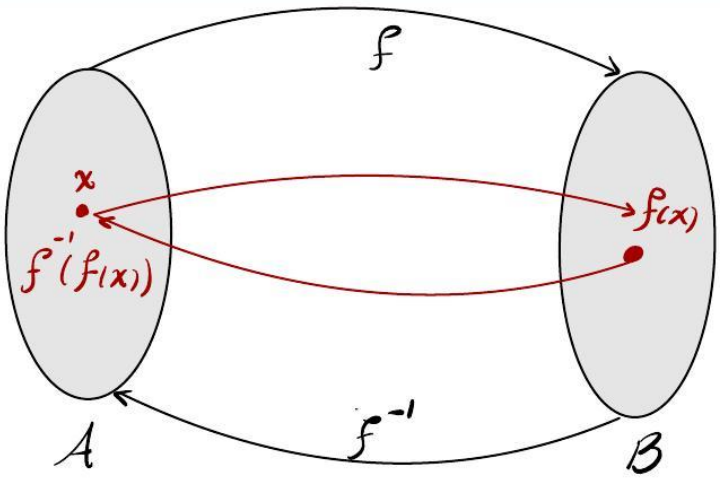
$f^{-1}(-10)=8$ b/c $f(8)=-10$.

$$f^{-1}(x)=y \iff f(y)=x$$

We have cancellation equations.

$f^{-1}(f(x))=x$ for every $x\in \text{Dom } f=A$.

$f(f^{-1}(x))=x$ for every $x\in \text{Range } f=B$.



For example if $f(x) = x^3$, $f^{-1}(x) = x^{1/3}$

$$f^{-1}(f(x)) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

$$f(f^{-1}(x)) = f(x^{1/3}) = (x^{1/3})^3 = x.$$

How to find the inverse function of a one-to-one function?

Step 1. write $y = f(x)$

Step 2. Solve the equation for x in terms of y (if possible).

Step 3. To express f^{-1} as a function of x interchange x and y the resulting equation is $y = f^{-1}(x)$.

Example. Find the inverse function of $f(x) = x^3 + 2$.

Sln. $f(x)$ is one-to-one.

Step 1. $y = f(x) \iff y = x^3 + 2$.

Step 2. $y = x^3 + 2$

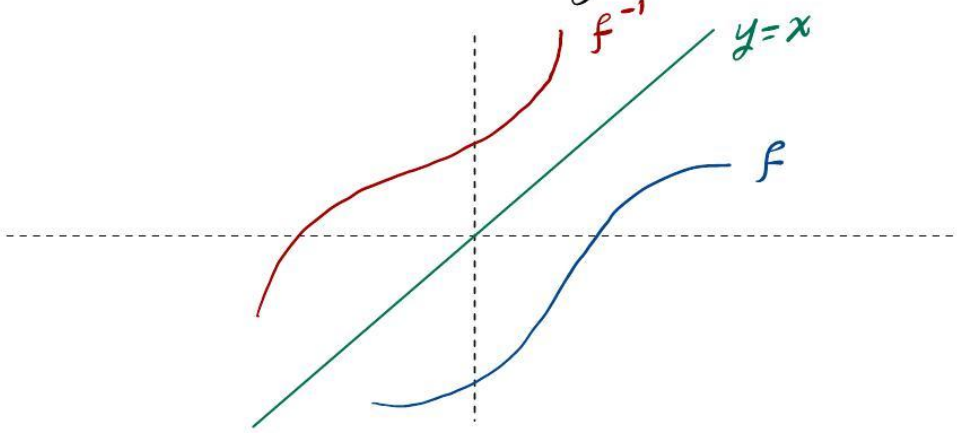
$$y - 2 = x^3$$

$$\sqrt[3]{y-2} = x.$$

Step 3. $y = \sqrt[3]{x-2}$

then $f^{-1}(x) = \sqrt[3]{x-2}$.

Notice that $\text{Dom } f^{-1} = \mathbb{R} = \text{Range } f$.



Rmk. The graph of f^{-1} is obtained by reflecting the graph of f about the line $y=x$.

* Derivatives of inverse functions.

Thm. If f is one-to-one continuous function defined on an interval, then the inverse f^{-1} is also continuous.

Thm. If f is one-to-one differentiable function with inverse f^{-1} and $f'(f^{-1}(a)) \neq 0$, then the inverse function is differentiable at a and

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}.$$

Pf.

We know that $f(f^{-1}(x)) = x$ for every x.

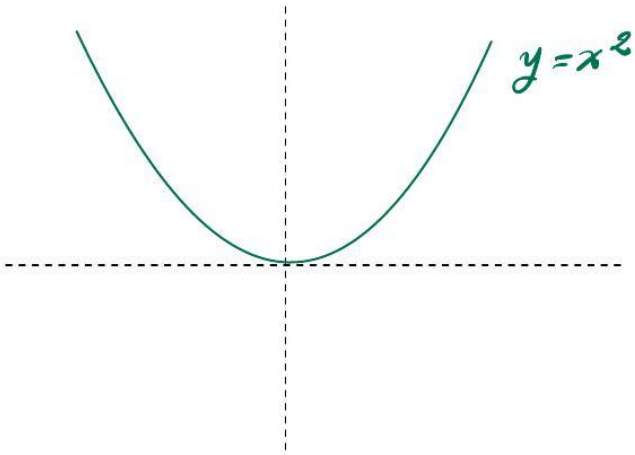
By implicit derivatives.

$$\frac{d}{dx}\left[f(f^{-1}(x))\right] = \frac{d}{dx}(x)$$

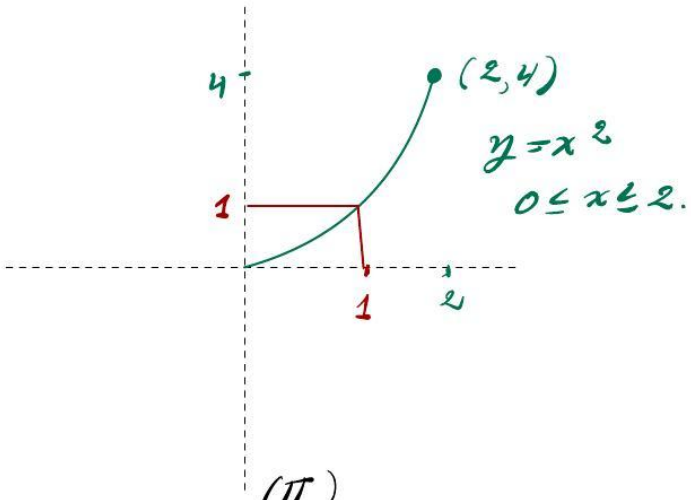
$$f'(f^{-1}(x)) \cdot \frac{d}{dx}(f^{-1}(x)) = 1$$

$$\frac{d}{dx}(f^{-1}(x)) = \frac{1}{f'(f^{-1}(x))}$$

Example Although the function $y=x^2$, $x \in \mathbb{R}$ is not one-to-one and therefore does not have an inverse, we can turn it into one-to-one function.



(I)



(II)

in the case (II) we have a one-to-one function. without computing a formula for $(f^{-1})'$ we can still calculate $(f^{-1})'(1)$

$f(1) = 1$ thus $f^{-1}(1) = 1$ since $f'(x) = 2x$ by the thm

$$(f^{-1})'(1) = \frac{1}{f'(f^{-1}(1))} = \frac{1}{f'(1)} = \frac{1}{2}.$$

for you: $(f^{-1})'(1)$ where $f(x) = 2x + \cos x$.